Theory of Statistics

Axiomatic Foundations, Proofs and Computational Applications

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Preface

This book aims to present the core ideas of mathematical statistics as a single whole, starting from measure-theoretic probability and continuing through point estimation, hypothesis testing, interval estimation and Bayesian inference. The final part unifies these topics within decision theory (minimaxity and admissibility) and extends them to three topics that arise often in practice: nonparametric methods, multiple testing and the false discovery rate, and the EM algorithm together with generalized linear models. Every chapter follows the same structure:

1. **Introduction and motivation:** the question the chapter sets out to answer.
2. **Definitions:** precise, axiomatic statements.
3. **Theorems and proofs:** the main results, with their assumptions stated explicitly.
4. **Worked examples:** what the theory says for the classical distributions.
5. **Computational applications:** Python code that checks the theoretical results by Monte Carlo simulation, together with the actual output of that code.
6. **Comments:** the limits of the results, common mistakes and links to other chapters.
7. **Chapter summary:** a box collecting the main results with their theorem numbers, and a note pointing to the corresponding chapters of standard textbooks.
8. **Exercises:** for consolidating the material.

Appendices. Appendix **??** at the end of the book is a reference table of the densities, means, variances and MGFs of common distributions, the relations between them and their conjugate priors. Appendix A gives solutions to all the exercises; the link at the end of each exercise leads to its solution, and the heading of each solution leads back to the exercise. The *Index* at the very end lists the pages on which the main concepts appear; the page on which a concept is defined is set in bold.

About the code. All code is written in Python 3 with `numpy`, `scipy` and `matplotlib`. Each code block is a separate file in the `code/` folder of the book (the file name is printed above the code box) and runs on its own; all of them can be run with `python3 code/run_all.py`. The simulations use numpy's recommended random number interface:

```
rng = np.random.default_rng(42)
```

Because the seed is fixed, the results are reproducible. The "Output" boxes directly below the code blocks were produced by actually running the code. Throughout the book the Gamma and exponential distributions are written with a *rate* parameter (see Notation), whereas `numpy` and `scipy` parametrise them by the *scale* $= 1/\text{rate}$. Expressions such as `scale=1/beta` in the code perform this conversion.

Notation

Symbol	**Meaning**
$(\Omega, \mathcal{F}, \mathbb{P})$	Probability space
$\mathcal{B}(\mathbb{R})$	Borel σ-algebra on $\mathbb{R}$
F_X, f_X, p_X	Distribution function (CDF), density (pdf), probability mass function (pmf)
$\mathbb{E}$, Var, Cov	Expectation, variance, covariance
$X_1, \ldots, X_n \overset{\text{i.i.d.}}{\sim} F$	Independent and identically distributed sample
$\bar{X}_n$, S_n^2	Sample mean, unbiased sample variance (divided by $n-1$)
$X_{(1)} \le \cdots \le X_{(n)}$	Order statistics
$\xrightarrow{P}$, $\xrightarrow{d}$, $\xrightarrow{\text{a.s.}}$, $\xrightarrow{L^p}$	Convergence in probability, in distribution, almost surely, in L^p
$L(\theta; \mathbf{x})$, $\ell(\theta; \mathbf{x})$	Likelihood and log-likelihood function
$I_1(\theta)$, $I_n(\theta)$	Fisher information for one and for n observations
Φ, φ, z_α	Standard normal CDF, pdf and upper α quantile: $\Phi(z_\alpha) = 1-\alpha$
$\chi^2_{k,\alpha}$, $t_{k,\alpha}$	Upper α quantiles of the chi-square and t distributions
Gamma(α, β)	Shape α, *rate* β: $f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$, mean α/β
Exp(λ)	Rate λ: Gamma$(1, \lambda)$, mean $1/\lambda$; also $\chi^2_k = \text{Gamma}(k/2, \frac{1}{2})$
$\mathbb{I}_A$	Indicator function of the event A
$\perp\!\!\!\perp$	Independence

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Part I

Foundations of Probability Theory

Chapter 1

Probability Spaces and Kolmogorov's Axioms

Probability theory is the mathematics of assigning numerical weights, in a consistent way, to the outcomes of experiments that involve uncertainty. The word "consistent" is crucial: unless we specify in advance which sets may receive a probability (the σ-algebra) and which rules the assignment must obey (Kolmogorov's axioms), arguments that look intuitively correct can lead to contradictions. In this chapter we lay the foundation on which the rest of the book is built.

1.1 Definitions

Definition 1.1 (Sample Space and Event)**.** The set of all possible outcomes of a random experiment is called the *sample space* and is denoted by Ω. Its elements $\omega \in \Omega$ are called *outcomes* (or elementary events), and the subsets $A \subseteq \Omega$ to which a probability can be assigned are called *events*.

Definition 1.2 (σ-Algebra)**.** A family $\mathcal{F} \subseteq \mathcal{P}(\Omega)$ of subsets of the sample space Ω is a σ-algebra if

1. $\Omega \in \mathcal{F}$,
2. $A \in \mathcal{F} \implies A^c = \Omega \setminus A \in \mathcal{F}$,
3. $A_1, A_2, \dots \in \mathcal{F} \implies \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$.

The pair $(\Omega, \mathcal{F})$ is called a *measurable space*. If the third condition is required only for finite unions, $\mathcal{F}$ is called an *algebra* (or field).

Remark 1.3. By De Morgan's law $\left(\bigcup_i A_i\right)^c = \bigcap_i A_i^c$, a σ-algebra is also closed under countable intersections, under differences ($A \setminus B = A \cap B^c$), and contains $\emptyset = \Omega^c$.

Definition 1.4 (Generated σ-Algebra)**.** Let $\mathcal{C} \subseteq \mathcal{P}(\Omega)$ be any family of sets. The intersection of all σ-algebras containing $\mathcal{C}$,

$$\sigma(\mathcal{C}) = \bigcap \{\mathcal{G} : \mathcal{G} \text{ is a } \sigma\text{-algebra and } \mathcal{C} \subseteq \mathcal{G}\},$$

is again a σ-algebra, and it is the *smallest* σ-algebra containing $\mathcal{C}$. (The intersection is over a non-empty family, since $\mathcal{P}(\Omega)$ always belongs to it.)

Definition 1.5 (Kolmogorov's Axioms of Probability [43])**.** A function $\mathbb{P} : \mathcal{F} \to [0, 1]$ defined on a measurable space $(\Omega, \mathcal{F})$ is a *probability measure* if it satisfies:

1. (Non-negativity) $\mathbb{P}(A) \geq 0$ for every $A \in \mathcal{F}$.
2. (Normalisation) $\mathbb{P}(\Omega) = 1$.
3. (Countable additivity) For every sequence $\{A_i\}_{i=1}^{\infty}$ of pairwise disjoint events, $\mathbb{P}\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.

The triple $(\Omega, \mathcal{F}, \mathbb{P})$ is called a *probability space*.

Definition 1.6 (Conditional Probability)**.** For $\mathbb{P}(B) > 0$, the conditional probability of A given B is defined by

$$\mathbb{P}(A \mid B) = \frac{\mathbb{P}(A \cap B)}{\mathbb{P}(B)}.$$

For fixed B, the map $A \mapsto \mathbb{P}(A \mid B)$ is a new probability measure on $(\Omega, \mathcal{F})$.

Definition 1.7 (Independence)**.** Events $A, B \in \mathcal{F}$ are *independent* if $\mathbb{P}(A \cap B) = \mathbb{P}(A)\mathbb{P}(B)$. A family $\{A_i\}_{i \in \mathcal{I}}$ is *mutually independent* if

$$\mathbb{P}\Big(\bigcap_{j \in J} A_j\Big) = \prod_{j \in J} \mathbb{P}(A_j)$$

for every finite $J \subseteq \mathcal{I}$. If the condition is required only for $|J| = 2$, the family is *pairwise independent*.

Definition 1.8 (Limit Superior and Limit Inferior of a Sequence of Events)**.** For a sequence of events $\{A_n\}$,

$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \{A_n \text{ occurs infinitely often}\}, \qquad \liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k.$$

$\liminf A_n$ is the event "from some index on, all the A_n occur". Both belong to $\mathcal{F}$.

1.2 Main Theorems and Proofs

Proposition 1.9 (Basic Properties of a Probability Measure)**.** *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $A, B \in \mathcal{F}$.*

1. $\mathbb{P}(\emptyset) = 0$.
2. *(Finite additivity) If $A_1, \dots, A_n$ are pairwise disjoint, then $\mathbb{P}(\bigcup_{i=1}^{n} A_i) = \sum_{i=1}^{n} \mathbb{P}(A_i)$.*
3. $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
4. *(Monotonicity) $A \subseteq B \implies \mathbb{P}(A) \leq \mathbb{P}(B)$ and $\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A)$.*
5. *(Inclusion–exclusion) $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.*
6. *(Boole's inequality) For every $\{A_i\} \subseteq \mathcal{F}$, $\mathbb{P}(\bigcup_{i=1}^{\infty} A_i) \leq \sum_{i=1}^{\infty} \mathbb{P}(A_i)$.*

Proof. (1) Applying Axiom 3 to the disjoint sequence $A_1 = \Omega$, $A_i = \emptyset$ $(i \geq 2)$ gives $1 = 1 + \sum_{i \geq 2} \mathbb{P}(\emptyset)$; a sum of non-negative terms equal to zero forces $\mathbb{P}(\emptyset) = 0$.

(2) Take $A_{n+1} = A_{n+2} = \cdots = \emptyset$ and use Axiom 3 together with (1).

(3) From the disjoint union $\Omega = A \cup A^c$ we get $1 = \mathbb{P}(A) + \mathbb{P}(A^c)$.

(4) $B = A \cup (B \setminus A)$ is a disjoint union; hence $\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A)$.

(5) From the disjoint unions $A \cup B = A \cup (B \setminus A)$ and $B = (A \cap B) \cup (B \setminus A)$ we obtain $\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B)$.

(6) The sets $B_1 = A_1$, $B_n = A_n \setminus \bigcup_{i<n} A_i$ are pairwise disjoint, $B_n \subseteq A_n$ and $\bigcup B_n = \bigcup A_n$. By Axiom 3 and monotonicity, $\mathbb{P}(\bigcup A_n) = \sum \mathbb{P}(B_n) \leq \sum \mathbb{P}(A_n)$. $\square$

Theorem 1.10 (Continuity of Probability)**.** *If $A_1 \subseteq A_2 \subseteq \dots$ is an increasing sequence of events and $A = \bigcup_{n=1}^{\infty} A_n$, then*

$$\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A).$$

Proof. Define disjoint rings: $C_1 = A_1$ and $C_n = A_n \setminus A_{n-1} \in \mathcal{F}$ for $n \geq 2$. The sets are pairwise disjoint, $\bigcup_{i=1}^{n} C_i = A_n$ and $\bigcup_{i=1}^{\infty} C_i = A$. By Kolmogorov's Axiom 3 and finite additivity,

$$\mathbb{P}(A) = \mathbb{P}\left(\bigcup_{i=1}^{\infty} C_i\right) = \sum_{i=1}^{\infty} \mathbb{P}(C_i) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{P}(C_i) = \lim_{n\to\infty} \mathbb{P}(A_n). \qquad \square$$

Corollary 1.11 (Continuity from Above)**.** *If $A_1 \supseteq A_2 \supseteq \dots$ is a decreasing sequence and $A = \bigcap_{n=1}^{\infty} A_n$, then $\lim_{n\to\infty} \mathbb{P}(A_n) = \mathbb{P}(A)$.*

Proof. The sequence A_n^c is increasing, and by De Morgan $\bigcup_n A_n^c = A^c$. By Theorem 1.10, $1 - \mathbb{P}(A_n) = \mathbb{P}(A_n^c) \to \mathbb{P}(A^c) = 1 - \mathbb{P}(A)$. $\square$

Remark 1.12. Theorem 1.10 is in fact *equivalent* to Axiom 3: every finitely additive set function that is continuous from below is countably additive. Continuity is thus the "analytic face" of countable additivity. In Chapter 2, the right-continuity of the CDF will follow directly from Corollary 1.11.

Theorem 1.13 (Law of Total Probability and Bayes' Formula)**.** *Let $\{B_i\}_{i\geq 1}$ be a countable partition of Ω (pairwise disjoint with $\bigcup B_i = \Omega$) with $\mathbb{P}(B_i) > 0$. For every $A \in \mathcal{F}$,*

$$\mathbb{P}(A) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i), \qquad \mathbb{P}(B_j \mid A) = \frac{\mathbb{P}(A \mid B_j)\mathbb{P}(B_j)}{\sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)} \quad (\mathbb{P}(A) > 0).$$

Proof. $A = \bigcup_i (A \cap B_i)$ is a disjoint union; by Axiom 3 and the definition of conditional probability, $\mathbb{P}(A) = \sum_i \mathbb{P}(A \cap B_i) = \sum_i \mathbb{P}(A \mid B_i)\mathbb{P}(B_i)$. Bayes' formula follows by writing the numerator and denominator of $\mathbb{P}(B_j \mid A) = \mathbb{P}(A \cap B_j)/\mathbb{P}(A)$ in this way. $\square$

Theorem 1.14 (Borel–Cantelli Lemmas; see [22, Section 2.3])**.** *Let $\{A_n\}$ be a sequence of events.*

1. *If $\sum_{n=1}^{\infty} \mathbb{P}(A_n) < \infty$, then $\mathbb{P}(\limsup A_n) = 0$.*
2. *If the $\{A_n\}$ are mutually independent and $\sum_{n=1}^{\infty} \mathbb{P}(A_n) = \infty$, then $\mathbb{P}(\limsup A_n) = 1$.*

Proof. (1) For every n, $\limsup A_k \subseteq \bigcup_{k\geq n} A_k$; by Boole's inequality $\mathbb{P}(\limsup A_k) \leq \sum_{k\geq n} \mathbb{P}(A_k)$. The right-hand side is the tail of a convergent series and tends to zero as $n \to \infty$.

(2) Since $(\limsup A_n)^c = \bigcup_n \bigcap_{k\geq n} A_k^c$, it suffices to show that $\mathbb{P}(\bigcap_{k\geq n} A_k^c) = 0$ for every n. By independence and the inequality $1 - x \leq e^{-x}$, for every $m > n$,

$$\mathbb{P}\Big(\bigcap_{k=n}^{m} A_k^c\Big) = \prod_{k=n}^{m} (1 - \mathbb{P}(A_k)) \leq \exp\Big(-\sum_{k=n}^{m} \mathbb{P}(A_k)\Big) \xrightarrow{m\to\infty} 0.$$

Corollary 1.11 then gives $\mathbb{P}(\bigcap_{k\geq n} A_k^c) = 0$. $\square$

1.3 Worked Examples

Example 1.15 (A Finite σ-Algebra)**.** For a roll of a die let $\Omega = \{1, \dots, 6\}$. If we are interested only in the information "odd or even?", then $\mathcal{F} = \{\emptyset, \{1,3,5\}, \{2,4,6\}, \Omega\}$ is a σ-algebra. This $\mathcal{F}$ does not contain the event $\{1\}$, so at this level of information no probability can be assigned to the event "a 1 was rolled". Interpreting a σ-algebra as a body of *information* is the basis of the conditional expectation in Chapter 3.

Example 1.16 (An Algebra That Is Not a σ-Algebra)**.** Let $\Omega = \mathbb{N}$ and $\mathcal{A} = \{A \subseteq \mathbb{N} : A \text{ is finite or } A^c \text{ is finite}\}$. $\mathcal{A}$ is closed under complements and finite unions, so it is an algebra. However, the set of even numbers, which is the union of the finite sets $A_i = \{2i\}$, is neither finite nor co-finite; hence $\mathcal{A}$ is not a σ-algebra.

Example 1.17 (The Birthday Problem)**.** Suppose the birthdays in a group of n people are independent and uniformly distributed over 365 days. By the complement rule, the probability that at least two people share a birthday is

$$p_n = 1 - \prod_{k=0}^{n-1} \frac{365 - k}{365}.$$

The approximation $1 - x \approx e^{-x}$ gives $p_n \approx 1 - \exp\{-n(n-1)/730\}$; for $n = 23$, $p_{23} \approx 0.507$. This counter-intuitive result is explained by the existence of $\binom{23}{2} = 253$ distinct pairs of people.

Example 1.18 (A Medical Diagnostic Test — Bayes' Formula)**.** Let the prevalence of a disease be $\mathbb{P}(D) = 0.01$, the sensitivity of the test $\mathbb{P}(+ \mid D) = 0.99$ and its specificity $\mathbb{P}(- \mid D^c) = 0.95$. By Theorem 1.13,

$$\mathbb{P}(D \mid +) = \frac{0.99 \cdot 0.01}{0.99 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.0099}{0.0594} \approx 0.167.$$

A person with a positive result is actually ill with probability only 16.7%. A low prior probability (prevalence) lowers the positive predictive value of even a highly accurate test. This is the simplest instance of the discussion of "the role of prior information" in Chapter **??**.

Example 1.19 (Events That Are Pairwise but Not Mutually Independent)**.** Toss two fair coins: $A = \{\text{first toss H}\}$, $B = \{\text{second toss H}\}$, $C = \{\text{both tosses equal}\}$. Since $\mathbb{P}(A) = \mathbb{P}(B) = \mathbb{P}(C) = 1/2$ and every pairwise intersection has probability $1/4$, the events are pairwise independent. However, $\mathbb{P}(A \cap B \cap C) = \mathbb{P}(\{HH\}) = 1/4 \neq 1/8 = \mathbb{P}(A)\mathbb{P}(B)\mathbb{P}(C)$. Mutual independence is therefore a strictly stronger condition than pairwise independence.

Example 1.20 (An Application of Borel–Cantelli)**.** If the events A_n are independent with $\mathbb{P}(A_n) = 1/n^2$, then since $\sum 1/n^2 = \pi^2/6 < \infty$, with probability 1 only finitely many of the A_n occur. If $\mathbb{P}(A_n) = 1/n$, then $\sum 1/n = \infty$ and, under independence, the A_n occur infinitely often with probability 1. The difference between the two sequences decides whether "rare" events eventually die out.

1.4 Computational Applications

1.4.1 Verifying Continuity by Monte Carlo

For $\Omega = [0,1]$ with Lebesgue measure $\mathbb{P}$ and the increasing sequence $A_n = [0, 1 - 1/n]$, we have $A = \bigcup A_n = [0,1)$ and $\mathbb{P}(A) = 1$. By Theorem 1.10, $\mathbb{P}(A_n) = 1 - 1/n \to 1$.

code/ch01/c01_1_continuity_monte_carlo.py

```python
import numpy as np

rng = np.random.default_rng(42)
N_SAMPLES = 500_000
omega_samples = rng.uniform(0.0, 1.0, size=N_SAMPLES)

for n in [1, 5, 10, 25, 50]:
    threshold = 1.0 - (1.0 / n)
    p_sim = np.mean(omega_samples <= threshold)
    print(f"n = {n:2d} | Theoretical P(A_n): {threshold:.4f} | Monte Carlo: {p_sim:.4f}")
```

Output

```
n =  1 | Theoretical P(A_n): 0.0000 | Monte Carlo: 0.0000
n =  5 | Theoretical P(A_n): 0.8000 | Monte Carlo: 0.8005
n = 10 | Theoretical P(A_n): 0.9000 | Monte Carlo: 0.9000
n = 25 | Theoretical P(A_n): 0.9600 | Monte Carlo: 0.9601
n = 50 | Theoretical P(A_n): 0.9800 | Monte Carlo: 0.9799
```

1.4.2 The Birthday Problem: Exact Value and Simulation

code/ch01/c01_2_birthday_problem.py

```python
import numpy as np

rng = np.random.default_rng(42)
def exact(n):
    return 1.0 - np.prod((365 - np.arange(n)) / 365)

R = 100_000
for n in [10, 23, 30, 50]:
    days = rng.integers(0, 365, size=(R, n))
    days.sort(axis=1)
    collision = np.any(np.diff(days, axis=1) == 0, axis=1)
    approx = 1 - np.exp(-n * (n - 1) / 730)
    print(f"n={n:2d} | exact={exact(n):.4f} | exponential approx.={approx:.4f} | MC={
    collision.mean():.4f}")
```

Output

```
n=10 | exact=0.1169 | exponential approx.=0.1160 | MC=0.1179
n=23 | exact=0.5073 | exponential approx.=0.5000 | MC=0.5067
n=30 | exact=0.7063 | exponential approx.=0.6963 | MC=0.7086
n=50 | exact=0.9704 | exponential approx.=0.9651 | MC=0.9707
```

1.4.3 Bayes' Formula: Simulating a Diagnostic Test

code/ch01/c01_3_bayes_diagnostic_test.py

```python
import numpy as np

rng = np.random.default_rng(42)
N = 2_000_000
ill = rng.random(N) < 0.01
positive = np.where(ill, rng.random(N) < 0.99, rng.random(N) < 0.05)
print(f"P(D | +) theory    : {0.0099/0.0594:.4f}")
print(f"P(D | +) simulation: {ill[positive].mean():.4f}")
print(f"P(+)     theory    : {0.0594:.4f} | simulation: {positive.mean():.4f}")
```

Output

```
P(D | +) theory    : 0.1667
P(D | +) simulation: 0.1664
P(+)     theory    : 0.0594 | simulation: 0.0592
```

1.5 Comments and Notes

Remark 1.21 (Why Not Assign a Probability to Every Subset?). It is impossible to assign a translation-invariant, countably additive "length" to *all* subsets of $\Omega = [0,1]$ (the Vitali set; this requires the axiom of choice). We therefore define probability only on a σ-algebra of "sufficiently regular" sets. Every set met in practice (intervals, their countable unions and intersections, and so on) belongs to the Borel σ-algebra.

Remark 1.22 (Probability Zero Is Not Impossibility). In the uniform example above, $\mathbb{P}(\{\omega\}) = 0$ for every single point $\{\omega\}$, yet the outcome of the experiment is always some point. This is why events of probability 1 are said to occur "almost surely" (a.s.).

Remark 1.23 (Monte Carlo Error). When a probability p is estimated by the proportion in N independent trials, the standard error is $\sqrt{p(1-p)/N}$. For $N = 500\,000$ and $p = 0.5$ this is about 0.0007; the differences between the theoretical and simulated values in the outputs above are of this order. The proportion is guaranteed to converge by the Law of Large Numbers in Chapter **??**.

1.6 Chapter Summary

- A probability model is a triple $(\Omega, \mathcal{F}, \mathbb{P})$: the events form a σ-algebra (Definition 1.2) and $\mathbb{P}$ satisfies Kolmogorov's axioms, in particular countable additivity (Definition 1.5).
- The complement rule, monotonicity, inclusion–exclusion and Boole's inequality are derived from the axioms (Proposition 1.9).
- A probability measure is continuous along monotone sequences of events (Theorem 1.10). Being able to interchange limits and probability underlies all later convergence results.
- Conditional probability, the law of total probability and Bayes' formula (Theorem 1.13) are the probabilistic counterpart of Bayesian inference (Chapter **??**). Mutual independence is stronger than pairwise independence.
- Borel–Cantelli lemmas (Theorem 1.14): if $\sum \mathbb{P}(A_n) < \infty$, the A_n occur infinitely often with probability 0; for independent events with $\sum \mathbb{P}(A_n) = \infty$ this probability is 1. They are the main tool in proofs of almost sure convergence.

Further reading. For measure-theoretic foundations see [22, Chapter 1] and [9, Sections 1–4]; for a more introductory treatment see [16, Chapter 1] and [81, Chapter 1].

1.7 Exercises

Exercise 1.1. Show that if $\mathcal{F}_1$ and $\mathcal{F}_2$ are σ-algebras, then so is $\mathcal{F}_1 \cap \mathcal{F}_2$, but that $\mathcal{F}_1 \cup \mathcal{F}_2$ in general is not, by giving an example. → Solution

Exercise 1.2. Prove the inclusion–exclusion formula for three events and obtain the general formula for n events by induction. Use it to show that when n letters are placed in envelopes at random, the probability that no letter goes into its own envelope converges to e^{-1} as $n \to \infty$. → Solution

Exercise 1.3. Prove the Bonferroni inequality $\mathbb{P}(\bigcap_{i=1}^{n} A_i) \geq 1 - \sum_{i=1}^{n} \mathbb{P}(A_i^c)$. Discuss its use in multiple hypothesis testing (Chapter **??**). → Solution

Exercise 1.4. Show that if A and B are independent, then so are A and B^c, and A^c and B^c. → Solution

Exercise 1.5. Write Python code that plots $\mathbb{P}(D \mid +)$ in Example 1.18 as the prevalence $\mathbb{P}(D)$ varies over $[0.001, 0.5]$. Find analytically the prevalence threshold above which the positive predictive value exceeds 50%. → Solution

Chapter 2

Random Variables and Distribution Functions

A probability space is an abstract structure: Ω may consist of sequences of coin tosses, particle positions or stock price paths. In statistics, however, we work with *numerical* summaries of these outcomes. A random variable is a function that carries the abstract Ω into the real numbers while preserving the probability structure. In this chapter we show that the distribution of a random variable is completely determined by a single function, the CDF; we study the characteristic properties of the CDF and the inverse transform method that underlies random number generation.

2.1 Definitions

Definition 2.1 (Borel σ-Algebra and Random Variable)**.** The smallest σ-algebra generated by all open intervals of $\mathbb{R}$ is called the Borel σ-algebra ($\mathcal{B}(\mathbb{R})$). A function from a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ to $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ such that $X^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{R})$ is called a *random variable* (RV).

Remark 2.2. $\mathcal{B}(\mathbb{R})$ is also generated by the family $\{(-\infty, x] : x \in \mathbb{R}\}$. Hence, for X to be a random variable it is *sufficient* that $\{\omega : X(\omega) \leq x\} \in \mathcal{F}$ for every $x \in \mathbb{R}$. (Idea of proof: the family $\{B : X^{-1}(B) \in \mathcal{F}\}$ is a σ-algebra containing the generating family.)

Definition 2.3 (Distribution)**.** If X is an RV, the measure $\mathbb{P}_X$ defined by $\mathbb{P}_X(B) = \mathbb{P}(X^{-1}(B)) = \mathbb{P}(X \in B)$, $B \in \mathcal{B}(\mathbb{R})$, is called the *distribution* (or law) of X. $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_X)$ is a probability space in its own right. If $\mathbb{P}_X = \mathbb{P}_Y$, then X and Y are said to be *identically distributed*, written $X \stackrel{d}{=} Y$.

Definition 2.4 (Distribution Function (CDF))**.** The cumulative distribution function of X, $F_X : \mathbb{R} \to [0, 1]$, is defined by $F_X(x) = \mathbb{P}(X \leq x)$.

Definition 2.5 (Discrete and Continuous Random Variables)**.**

1. X is *discrete* if it takes values in a countable set $\{x_1, x_2, \dots\}$; the function $p_X(x) = \mathbb{P}(X = x)$ is called the *probability mass function* (pmf), and $F_X(x) = \sum_{x_i \leq x} p_X(x_i)$.
2. X is *(absolutely) continuous* if $F_X(x) = \int_{-\infty}^{x} f_X(t)\, dt$ for some non-negative integrable f_X; f_X is called the *probability density function* (pdf). At points where f_X is continuous, $F_X'(x) = f_X(x)$.

Definition 2.6 (Quantile (Generalised Inverse) Function)**.** For $u \in (0, 1)$, the function $F^{-1}(u) = \inf\{x \in \mathbb{R} : F(x) \geq u\}$ is called the *quantile function* of F. If F is continuous and strictly increasing, this is the ordinary inverse. $F^{-1}(1/2)$ is the median.

Definition 2.7 (Empirical Distribution Function)**.** For a sample $X_1, \ldots, X_n$, the function

$$\hat{F}_n(x) = \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}\{X_i \leq x\}$$

is called the *empirical distribution function* (ECDF). For every fixed x, $n\hat{F}_n(x) \sim \text{Bin}(n, F(x))$.

2.2 Main Theorems and Proofs

Theorem 2.8 (Characteristic Properties of the CDF)**.** *Every distribution function F_X*

1. *is non-decreasing: $x < y \implies F_X(x) \leq F_X(y)$;*
2. *satisfies $\lim_{x \to -\infty} F_X(x) = 0$ and $\lim_{x \to \infty} F_X(x) = 1$;*
3. *is right-continuous.*

Conversely, every function F with these three properties is the CDF of some random variable on some probability space.

Proof. (1) If $x < y$ then $\{X \leq x\} \subseteq \{X \leq y\}$; apply the monotonicity in Proposition 1.9. (2) For $x_n \searrow -\infty$ the events $\{X \leq x_n\}$ decrease to $\emptyset$; by Corollary 1.11, $F_X(x_n) \to 0$. Similarly, for $x_n \nearrow \infty$ we have $\{X \leq x_n\} \nearrow \Omega$, and Theorem 1.10 gives $F_X(x_n) \to 1$. (3) This is Theorem 2.9 below. For the converse take $\Omega = (0,1)$, $\mathbb{P} =$ Lebesgue measure and $X(u) = F^{-1}(u)$; Theorem 2.13 shows that the CDF of this X is F. □

Theorem 2.9 (Right-Continuity of the CDF)**.** *The distribution function $F_X(x) = \mathbb{P}(X \leq x)$ is always right-continuous:*

$$\lim_{h \to 0^+} F_X(x + h) = F_X(x).$$

Proof. Since F_X is monotone, the right limit exists; it suffices to compute it along some sequence $h_n \searrow 0$. The events $A_n = \{\omega : X(\omega) \leq x + h_n\}$ form a decreasing chain $A_1 \supseteq A_2 \supseteq \ldots$ with $\bigcap_{n=1}^{\infty} A_n = \{\omega : X(\omega) \leq x\}$. By continuity of the measure from above (Corollary 1.11), $\lim_{n \to \infty} \mathbb{P}(A_n) = \mathbb{P}\left(\bigcap_{n=1}^{\infty} A_n\right) = F_X(x)$. □

Corollary 2.10 (Left Limits and Jumps)**.** *$F_X(x^-) = \lim_{h \to 0^+} F_X(x - h) = \mathbb{P}(X < x)$, and therefore $\mathbb{P}(X = x) = F_X(x) - F_X(x^-)$. That is, F_X jumps exactly at the points to which X assigns positive probability; for a continuous RV, $\mathbb{P}(X = x) = 0$ for every x.*

Proof. For $x - h_n \nearrow x$ we have $\{X \leq x - h_n\} \nearrow \{X < x\}$; apply Theorem 1.10. □

Proposition 2.11. *The set of discontinuity (jump) points of any CDF is at most countable.*

Proof. Let $D_k = \{x : F(x) - F(x^-) > 1/k\}$. Since the jumps cannot add up to more than 1, $|D_k| < k$. The set of all discontinuities is $\bigcup_{k \geq 1} D_k$, a countable union of finite sets, hence countable. □

Lemma 2.12 (Basic Equivalence for the Quantile Function)**.** *For every $u \in (0,1)$ and $x \in \mathbb{R}$, $F^{-1}(u) \leq x \iff u \leq F(x)$.*

Proof. ($\Leftarrow$) If $u \leq F(x)$ then $x \in \{z : F(z) \geq u\}$, hence the infimum satisfies $F^{-1}(u) \leq x$. ($\Rightarrow$) Since F is non-decreasing, the set $\{z : F(z) \geq u\}$ is an interval extending to $+\infty$; since F is right-continuous, the infimum belongs to the set, i.e. $F(F^{-1}(u)) \geq u$. If $F^{-1}(u) \leq x$, monotonicity gives $F(x) \geq F(F^{-1}(u)) \geq u$. □

Theorem 2.13 (Inverse Transform Method)**.** *If $U \sim$ Uniform$(0,1)$ and F is any CDF, then the random variable $X = F^{-1}(U)$ has CDF F.*

Proof. By Lemma 2.12, $\mathbb{P}(X \le x) = \mathbb{P}(F^{-1}(U) \le x) = \mathbb{P}(U \le F(x)) = F(x)$. □

Theorem 2.14 (Probability Integral Transform)**.** *If the CDF F of X is* continuous*, then $F(X) \sim$ Uniform$(0,1)$.*

Proof. Let $u \in (0,1)$. Negating Lemma 2.12 gives $\{F(X) < u\} = \{X < F^{-1}(u)\}$, so $\mathbb{P}(F(X) < u) = F(F^{-1}(u)^-)$. Since F is continuous, the left limit equals $F(F^{-1}(u))$, and by the intermediate value theorem $F(F^{-1}(u)) = u$. Thus $\mathbb{P}(F(X) < u) = u$ for every u; taking right limits gives $\mathbb{P}(F(X) \le u) = u$. □

Remark 2.15. Continuity is essential in Theorem 2.14: for $X \sim$ Bernoulli$(1/2)$, $F(X)$ takes the values $\{1/2, 1\}$ and is not uniformly distributed. The theorem shows that for continuous test statistics, p-values are uniformly distributed under H_0 (Chapter **??**).

Theorem 2.16 (Density Under a Monotone Transformation)**.** *Let X have density f_X, and let g be strictly monotone and differentiable on the support of X. The density of $Y = g(X)$ is*

$$f_Y(y) = f_X(g^{-1}(y)) \left| \frac{d}{dy} g^{-1}(y) \right|$$

(y in the range of g).

Proof. If g is increasing, $F_Y(y) = \mathbb{P}(g(X) \le y) = \mathbb{P}(X \le g^{-1}(y)) = F_X(g^{-1}(y))$; differentiate by the chain rule. If g is decreasing, $F_Y(y) = 1 - F_X(g^{-1}(y))$, and the minus sign of the derivative is absorbed by the absolute value. □

Theorem 2.17 (Glivenko–Cantelli [32, 15])**.** *If $X_1, X_2, \ldots \overset{i.i.d.}{\sim} F$, then*

$$D_n = \sup_{x \in \mathbb{R}} |\hat{F}_n(x) - F(x)| \xrightarrow{a.s.} 0.$$

Moreover, the Dvoretzky–Kiefer–Wolfowitz (DKW) inequality [21, 56] states that $\mathbb{P}(D_n > \varepsilon) \le 2e^{-2n\varepsilon^2}$ for every $\varepsilon > 0$.

Proof (first part). For fixed x, $\hat{F}_n(x) = \frac{1}{n}\sum_i \mathbb{I}\{X_i \le x\}$ and $\hat{F}_n(x^-) = \frac{1}{n}\sum_i \mathbb{I}\{X_i < x\}$ are averages of i.i.d. Bernoulli variables with means $F(x)$ and $F(x^-)$; by the strong law of large numbers (Theorem **??**) they converge a.s. to $F(x)$ and $F(x^-)$. The difficulty is taking the supremum over infinitely many x; we overcome it with a finite grid of quantiles.

Fix $k \in \mathbb{N}$. For $j = 1, \ldots, k-1$ let $x_j = F^{-1}(j/k) = \inf\{x : F(x) \ge j/k\}$, and set $x_0 = -\infty$, $x_k = +\infty$. By the definition of the quantile function, $F(x_j^-) \le j/k \le F(x_j)$, hence $F(x_j^-) - F(x_{j-1}) \le \frac{1}{k}$. For $x \in [x_{j-1}, x_j)$, monotonicity of $\hat{F}_n$ and F gives

$$\hat{F}_n(x) - F(x) \le \hat{F}_n(x_j^-) - F(x_{j-1}) \le \hat{F}_n(x_j^-) - F(x_j^-) + \tfrac{1}{k},$$

$$\hat{F}_n(x) - F(x) \ge \hat{F}_n(x_{j-1}) - F(x_j^-) \ge \hat{F}_n(x_{j-1}) - F(x_{j-1}) - \tfrac{1}{k}.$$

Since these intervals cover $\mathbb{R}$,

$$D_n \le \max_{0 \le j \le k} \left\{ |\hat{F}_n(x_j) - F(x_j)|,\ |\hat{F}_n(x_j^-) - F(x_j^-)| \right\} + \frac{1}{k}$$

($\hat{F}_n$ and F agree at the end points). The maximum on the right has finitely many terms, each tending to zero a.s.; hence on an event Ω_k of probability 1, $\limsup_n D_n \le 1/k$. On $\bigcap_k \Omega_k$ we have $D_n \to 0$, and this intersection has probability 1. The proof of the DKW inequality requires a finer combinatorial argument; for the sharp constant 2 see [56]. □

Remark 2.18. The Glivenko–Cantelli theorem is often called "the fundamental theorem of statistics": a large enough sample reveals the whole unknown distribution uniformly. The limit distribution of $\sqrt{n}D_n$ is the Kolmogorov distribution, which is the basis of the Kolmogorov–Smirnov goodness-of-fit test. Pointwise convergence ($\hat{F}_n(x) \xrightarrow{\text{a.s.}} F(x)$) follows at once from the law of large numbers in Chapter **??**; the strength of the theorem lies in the *uniform* convergence.

2.3 Worked Examples

Example 2.19 (The Bernoulli CDF)**.** For $X \sim \text{Bernoulli}(p)$,

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 1-p, & 0 \le x < 1 \\ 1, & x \ge 1 \end{cases}$$

The function jumps at 0 and 1, with jump sizes $1-p$ and p respectively. At each jump F_X is right-continuous but not left-continuous.

Example 2.20 (A Mixed Distribution: Insurance Claims)**.** Suppose that a policy has no claim during the year with probability 0.7 and that, given a claim, the amount is $\text{Exp}(\lambda)$-distributed. The CDF of the claim amount X is

$$F_X(x) = \begin{cases} 0, & x < 0 \\ 0.7 + 0.3(1 - e^{-\lambda x}), & x \ge 0 \end{cases}$$

X is neither discrete nor continuous: it has a jump of size 0.7 at 0 and density $0.3\lambda e^{-\lambda x}$ on $(0, \infty)$. Such distributions show why the CDF approach is more general than the pmf/pdf approach.

Example 2.21 (Deriving the Chi-Square Distribution)**.** Let $X \sim \mathcal{N}(0,1)$ and $Y = X^2$. Since $g(x) = x^2$ is not monotone, we use the CDF method: for $y > 0$,

$$F_Y(y) = \mathbb{P}(-\sqrt{y} \le X \le \sqrt{y}) = 2\Phi(\sqrt{y}) - 1, \qquad f_Y(y) = \frac{2\varphi(\sqrt{y})}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi y}} e^{-y/2}.$$

This is the χ^2_1 density. It will play a central role for confidence intervals in Chapter **??**.

Example 2.22 (Generating Exponential and Cauchy Variates by Inversion)**.** For $F(x) = 1 - e^{-\lambda x}$ $(x \ge 0)$, $F^{-1}(u) = -\ln(1-u)/\lambda$; hence if $U \sim \text{Uniform}(0,1)$, then $X = -\ln(1-U)/\lambda \sim \text{Exp}(\lambda)$. (Since $1 - U \overset{d}{=} U$, $-\ln U/\lambda$ can also be used.) For the standard Cauchy, $F(x) = \frac{1}{2} + \frac{1}{\pi}\arctan x$ and $F^{-1}(u) = \tan\left(\pi(u - \tfrac{1}{2})\right)$.

Example 2.23 (Inverse Transform for a Discrete Distribution)**.** If $\mathbb{P}(X = k) = p_k$, $k = 0, 1, 2, \ldots$, then $F^{-1}(u) = \min\{k : \sum_{j \le k} p_j \ge u\}$. That is, generate U and return the first index at which the cumulative probabilities exceed U. This is a concrete example showing that Theorem 2.13 also holds for discrete distributions, and it explains why the generalised inverse is defined with an inf.

2.4 Computational Applications

2.4.1 Empirical CDF and the Kolmogorov–Smirnov Distance

code/ch02/c02_1_ecdf_ks_distance.py

```python
import numpy as np
from scipy.stats import norm

rng = np.random.default_rng(42)
x_grid = np.linspace(-3.5, 3.5, 1000)
f_theo = norm.cdf(x_grid)

for n in [20, 100, 1000, 10000]:
    samples = rng.normal(0.0, 1.0, size=n)
    ecdf = np.array([np.mean(samples <= x) for x in x_grid])
    ks_dist = np.max(np.abs(ecdf - f_theo))
    print(f"n = {n:5d} | KS distance: {ks_dist:.4f}")
```

Output

```
n =    20 | KS distance: 0.1290
n =   100 | KS distance: 0.1215
n =  1000 | KS distance: 0.0291
n = 10000 | KS distance: 0.0087
```

Note that the distance shrinks at a rate of roughly $1/\sqrt{n}$. The supremum computed over a grid is a lower approximation to the true D_n; the exact value can be computed in $O(n \log n)$ time from the sorted sample using $D_n = \max_i \max\{i/n - F(X_{(i)}),\, F(X_{(i)}) - (i-1)/n\}$. The code below uses this exact formula, compares it with the DKW bound and plots the ECDF.

code/ch02/c02_2_exact_ks_dkw_ecdf_plot.py

```python
import numpy as np
from scipy.stats import norm
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
def ks_exact(x, cdf):
    x = np.sort(x); n = len(x); F = cdf(x); i = np.arange(1, n + 1)
    return max(np.max(i / n - F), np.max(F - (i - 1) / n))

eps, R = 0.05, 20_000
for n in [100, 400, 1000]:
    D = np.array([ks_exact(rng.normal(size=n), norm.cdf) for _ in range(R)])
    print(f"n={n:4d} | P(D_n > {eps}) MC = {np.mean(D > eps):.4f} | DKW bound = {min(1, 2*np.exp(-2*n*eps**2)):.4f}"
          f" | E[sqrt(n) D_n] = {np.sqrt(n)*D.mean():.3f}")

fig, ax = plt.subplots(figsize=(7, 3.6))
xg = np.linspace(-3.5, 3.5, 500)
ax.plot(xg, norm.cdf(xg), "k-", lw=2, label="True F (N(0,1))")
for n, c in [(20, "tab:red"), (200, "tab:blue")]:
    s = np.sort(rng.normal(size=n))
    ax.step(np.r_[-3.5, s, 3.5], np.r_[0, np.arange(1, n + 1) / n, 1], where="post", color=c,
     label=f"ECDF, n={n}")
ax.set_xlabel("x"); ax.set_ylabel("F(x)"); ax.legend(); fig.tight_layout()
fig.savefig("ecdf.pdf")
```

Output

```
n= 100 | P(D_n > 0.05) MC = 0.9544 | DKW bound = 1.0000 | E[sqrt(n) D_n] = 0.851
n= 400 | P(D_n > 0.05) MC = 0.2600 | DKW bound = 0.2707 | E[sqrt(n) D_n] = 0.861
n=1000 | P(D_n > 0.05) MC = 0.0131 | DKW bound = 0.0135 | E[sqrt(n) D_n] = 0.863
```

The output above shows that $\mathbb{E}[\sqrt{n} D_n]$ stays roughly constant in n; this is a numerical confirmation of the $D_n = O_P(n^{-1/2})$ rate and of the Kolmogorov limit distribution (whose mean is $\sqrt{\pi/2}\ln 2 \approx 0.869$).

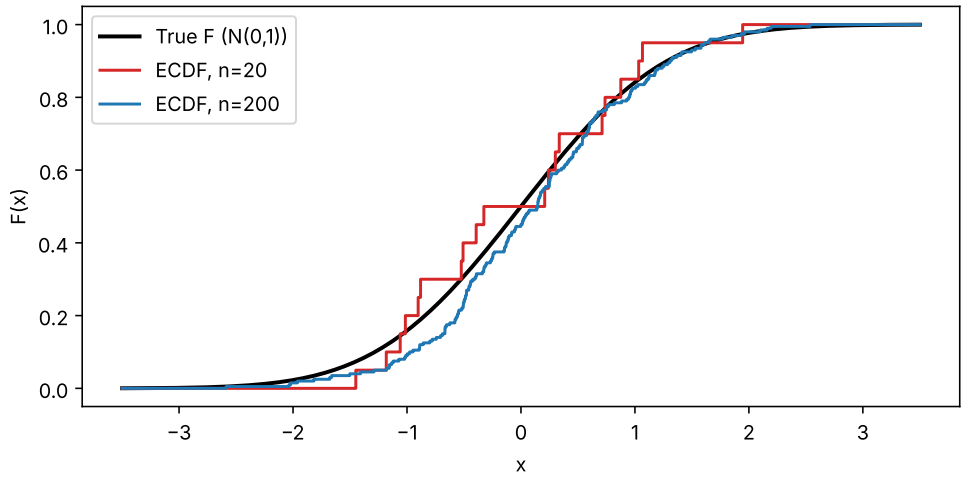


Figure 2.1: Empirical CDFs of standard normal samples. As n grows, the step function converges uniformly to the true CDF (Glivenko–Cantelli).

2.4.2 Inverse Transform and the Probability Integral Transform

code/ch02/c02_3_inverse_transform_pit.py

```python
import numpy as np
from scipy import stats

rng = np.random.default_rng(42)
U = rng.uniform(size=200_000)
lam = 2.0
X_exp = -np.log(1 - U) / lam                # Exp(2)
X_cau = np.tan(np.pi * (U - 0.5))           # standard Cauchy
X_poi = np.searchsorted(stats.poisson.cdf(np.arange(60), 3.0), U)  # Poisson(3)

print(f"Exponential: mean={X_exp.mean():.4f} (theory {1/lam}), KS p-value={stats.kstest(X_exp, 'expon', args=(0, 1/lam)).pvalue:.3f}")
print(f"Cauchy     : median={np.median(X_cau):.4f} (theory 0), KS p-value={stats.kstest(X_cau, 'cauchy').pvalue:.3f}")
print(f"Poisson    : mean={X_poi.mean():.4f}, variance={X_poi.var():.4f} (theory 3, 3)")

# Probability integral transform: F(X) ~ U(0,1) (for continuous F)
Z = rng.gamma(3.0, 2.0, size=200_000)      # Gamma(3, rate 1/2); numpy takes scale = 1/rate
V = stats.gamma.cdf(Z, a=3.0, scale=2.0)
print(f"PIT: mean of F(X)={V.mean():.4f} (1/2), variance={V.var():.4f} (1/12={1/12:.4f})")
```

Output

```
Exponential: mean=0.5002 (theory 0.5), KS p-value=0.964
Cauchy     : median=0.0004 (theory 0), KS p-value=0.964
Poisson    : mean=2.9998, variance=2.9995 (theory 3, 3)
PIT: mean of F(X)=0.4998 (1/2), variance=0.0833 (1/12=0.0833)
```

It is no coincidence that the KS p-values for the exponential and Cauchy samples are *exactly equal*: both samples were generated from the same vector U through their own F^{-1} and tested with their own F. Since $F(F^{-1}(U)) = U$, both tests actually test the same sample U against the uniform distribution. This is a direct illustration of the fact that the KS statistic is *distribution-free* for continuous F (Theorem 2.14).

2.5 Comments and Notes

Remark 2.24 (Why Is the CDF the Basic Object?). The distribution $\mathbb{P}_X$ of a random variable is a set function on $\mathcal{B}(\mathbb{R})$ and is hard to work with directly. By the Carathéodory extension theorem, $\mathbb{P}_X$ is *uniquely* determined by its values on the intervals $(-\infty, x]$, that is, by F_X. Hence the statement "two RVs have the same distribution" is equivalent to "their CDFs are equal".

Remark 2.25 (A Density Is Not a Probability). $f_X(x)$ is not a probability but a probability *density*; it may exceed 1. For example, for Uniform$(0, 1/2)$, $f_X \equiv 2$. Interpretation: for small h, $\mathbb{P}(x < X \le x + h) \approx f_X(x)h$.

Remark 2.26 (A Computational Note). The expression `[np.mean(samples <= x) for x in x_grid]` in the first code performs $O(n \cdot |\text{grid}|)$ operations. For large n, sorting the sample once and using `np.searchsorted` gives the ECDF in $O(n \log n)$ time.

2.6 Chapter Summary

- A random variable is a measurable function $X : \Omega \to \mathbb{R}$ (Definition 2.1). Its distribution is completely determined by its CDF.
- A function is a CDF if and only if it is non-decreasing and right-continuous with limits 0 and 1 (Theorem 2.8). Its jump points are at most countable, and $\mathbb{P}(X = x) = F(x) - F(x^-)$.
- The basic equivalence for the quantile function, $F^{-1}(u) \le x \iff u \le F(x)$ (Lemma 2.12), yields two results: the inverse transform method (Theorem 2.13) and, for continuous F, $F(X) \sim \text{Uniform}(0, 1)$ (Theorem 2.14).
- Under monotone transformations the density becomes $f_X(g^{-1}(y))\,|(g^{-1})'(y)|$ (Theorem 2.16).
- By the Glivenko–Cantelli theorem the empirical CDF converges uniformly to the true CDF (Theorem 2.17). The DKW inequality gives this convergence an exponential rate. The bootstrap and the Kolmogorov–Smirnov test rest on this result.

Further reading. [16, Chapters 1–2], [81, Chapter 2]; for the empirical CDF and the DKW inequality see [81, Chapter 7] and [78, Chapter 19].

2.7 Exercises

Exercise 2.1. If $X \sim \text{Uniform}(0, 1)$, find the distribution of $Y = -2 \ln X$ using Theorem 2.16 and show that it is χ^2_2. Relate this result to Fisher's method of combining p-values. → Solution

Exercise 2.2. Write the inverse transform algorithm for $F(x) = 1 - e^{-x^k}$ ($x \ge 0$, $k > 0$; Weibull) and compare the mean of your generated sample with $\Gamma(1 + 1/k)$. → Solution

Exercise 2.3. Let X and Y be independent with CDFs F and G. Find the CDFs of $\max(X, Y)$ and $\min(X, Y)$. Show that for $X_1, \dots, X_n \overset{\text{i.i.d.}}{\sim} F$ the CDF of $X_{(n)}$ is F^n (this will be used in Chapter **??**; for general order statistics see Theorem **??**). → Solution

Exercise 2.4. Using the DKW inequality, find the smallest n such that $\mathbb{P}(D_n > 0.02) \le 0.05$. From this, construct a 95% confidence *band* for F. → Solution

Chapter 3

Expectation, Variance and Cauchy–Schwarz

Knowing a whole distribution is often unnecessary or impossible; instead we work with summary numbers such as "centre" and "spread". The expectation is an integral with respect to the probability measure, and it is the common language of estimation theory (unbiasedness, MSE), limit theorems (LLN, CLT) and Bayesian decisions (posterior expected loss). In this chapter we discuss the measure-theoretic definition of expectation, the basic inequalities and the notion of conditional expectation, which will be used throughout the rest of the book.

3.1 Definitions

Definition 3.1 (Expectation as a Lebesgue Integral)**.** The expectation is defined in three steps:

1. *Simple RV:* for $X = \sum_{i=1}^{k} a_i \mathbb{I}_{A_i}$ ($A_i \in \mathcal{F}$ disjoint), $\mathbb{E}[X] = \sum_{i=1}^{k} a_i \mathbb{P}(A_i)$.
2. *Non-negative RV:* $\mathbb{E}[X] = \sup\{\mathbb{E}[Y] : Y \text{ simple},\ 0 \le Y \le X\}$ (the value may be $+\infty$).
3. *General RV:* write $X = X^+ - X^-$ ($X^+ = \max(X, 0)$, $X^- = \max(-X, 0)$); if at least one of $\mathbb{E}[X^+]$ and $\mathbb{E}[X^-]$ is finite, $\mathbb{E}[X] = \mathbb{E}[X^+] - \mathbb{E}[X^-]$. If $\mathbb{E}|X| < \infty$, X is called *integrable* ($X \in L^1$).

The result of these three steps is denoted by $\mathbb{E}[X] = \int_\Omega X \, d\mathbb{P}$.

Remark 3.2 (Computational Formulas). By the change-of-variables formula known as the Law of the Unconscious Statistician (LOTUS), for measurable g

$$\mathbb{E}[g(X)] = \int_{\mathbb{R}} g(x) \, d\mathbb{P}_X(x) = \begin{cases} \sum_x g(x) p_X(x), & X \text{ discrete} \\ \int_{-\infty}^{\infty} g(x) f_X(x) \, dx, & X \text{ continuous} \end{cases}$$

(provided the sum/integral converges absolutely). For $X \ge 0$ the tail formula $\mathbb{E}[X] = \int_0^\infty \mathbb{P}(X > t) \, dt$ also holds.

Definition 3.3 (Moments, Variance, Covariance, Correlation)**.** For $k \in \mathbb{N}$, $\mu'_k = \mathbb{E}[X^k]$ is the *k-th raw moment* and $\mu_k = \mathbb{E}[(X - \mathbb{E}X)^k]$ the *k-th central moment.* Furthermore,

$$\operatorname{Var}(X) = \mathbb{E}[(X - \mathbb{E}X)^2] = \mathbb{E}[X^2] - (\mathbb{E}X)^2,$$
$$\operatorname{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)] = \mathbb{E}[XY] - \mathbb{E}X \, \mathbb{E}Y,$$

$$\operatorname{Corr}(X, Y) = \rho_{XY} = \frac{\operatorname{Cov}(X, Y)}{\sqrt{\operatorname{Var}(X) \operatorname{Var}(Y)}}.$$

If $\operatorname{Cov}(X,Y)=0$, X and Y are called *uncorrelated*. The skewness is $\gamma_1=\mu_3/\sigma^3$ and the excess kurtosis is $\gamma_2=\mu_4/\sigma^4-3$.

Definition 3.4 (Moment Generating Function)**.** If $M_X(t)=\mathbb{E}[e^{tX}]$ is finite in a neighbourhood of 0, it is called the *moment generating function* (MGF) of X. In that case $\mathbb{E}[X^k]=M_X^{(k)}(0)$, and the MGF determines the distribution uniquely. (Not every distribution has an MGF; this gap will be filled by the characteristic function in Chapter **??**.)

Definition 3.5 (Conditional Expectation — the Elementary Case)**.** Let (X,Y) be a random vector with joint density (or pmf) $f_{X,Y}$, and let $f_Y(y)>0$. The conditional density of X given $Y=y$ is $f_{X\mid Y}(x\mid y)=f_{X,Y}(x,y)/f_Y(y)$, and with

$$m(y)=\mathbb{E}[X\mid Y=y]=\int x\, f_{X\mid Y}(x\mid y)\,dx$$

the *random variable* $\mathbb{E}[X\mid Y]=m(Y)$ is called the conditional expectation of X given Y. Similarly, $\operatorname{Var}(X\mid Y)=\mathbb{E}[(X-\mathbb{E}[X\mid Y])^2\mid Y]$.

Remark 3.6 (The General Definition)*.* In measure-theoretic terms, $\mathbb{E}[X\mid\mathcal{G}]$ is the (a.s. unique) RV that is measurable with respect to a sub-σ-algebra $\mathcal{G}\subseteq\mathcal{F}$ and satisfies $\mathbb{E}[X\mathbb{I}_G]=\mathbb{E}[\mathbb{E}[X\mid\mathcal{G}]\mathbb{I}_G]$ for every $G\in\mathcal{G}$; its existence follows from the Radon–Nikodym theorem. $\mathbb{E}[X\mid Y]=\mathbb{E}[X\mid\sigma(Y)]$. In L^2, $\mathbb{E}[X\mid\mathcal{G}]$ is the *orthogonal projection* of X onto the subspace of $\mathcal{G}$-measurable RVs; this geometric interpretation is the essence of the Rao–Blackwell theorem (Chapter **??**).

3.2 Main Theorems and Proofs

Proposition 3.7 (Properties of Expectation)**.** *Let $X,Y\in L^1$ and $a,b\in\mathbb{R}$.*

1. *(Linearity)* $\mathbb{E}[aX+bY]=a\mathbb{E}X+b\mathbb{E}Y$.
2. *(Monotonicity) If $X\le Y$ a.s., then $\mathbb{E}X\le\mathbb{E}Y$.*
3. *If $X\perp\!\!\!\perp Y$, then $\mathbb{E}[XY]=\mathbb{E}X\,\mathbb{E}Y$; hence independent RVs are uncorrelated.*
4. $\operatorname{Var}(aX+b)=a^2\operatorname{Var}(X)$ *and* $\operatorname{Var}(X+Y)=\operatorname{Var}X+\operatorname{Var}Y+2\operatorname{Cov}(X,Y)$.
5. *In general,* $\operatorname{Var}\left(\sum_{i=1}^n X_i\right)=\sum_i\operatorname{Var}X_i+2\sum_{i<j}\operatorname{Cov}(X_i,X_j)$.

Proof sketch. (1)–(2) follow directly for simple RVs and, in general, by passing to the limit with the Monotone Convergence Theorem. (3) First for indicators, $\mathbb{E}[\mathbb{I}_A\mathbb{I}_B]=\mathbb{P}(A\cap B)=\mathbb{P}(A)\mathbb{P}(B)$, then by linearity and a limiting argument. (4)–(5) follow by expanding the definition and using linearity. For measure-theoretic details see [22, Sections 1.4–1.6] or [9, Sections 15–16 and 21]. □

Theorem 3.8 (Cauchy–Schwarz Inequality)**.** *For any two random variables with $\mathbb{E}[X^2]<\infty$ and $\mathbb{E}[Y^2]<\infty$,*

$$(\mathbb{E}[XY])^2\le\mathbb{E}[X^2]\,\mathbb{E}[Y^2].$$

Equality holds if and only if $aX+bY=0$ a.s. for some $(a,b)\neq(0,0)$.

Proof. First, $\mathbb{E}[XY]$ is finite because $|XY|\le(X^2+Y^2)/2$. If $\mathbb{E}[X^2]=0$, then $X=0$ a.s. and both sides vanish. Let $\mathbb{E}[X^2]>0$. For every $t\in\mathbb{R}$,

$$g(t)=\mathbb{E}[(tX+Y)^2]=t^2\mathbb{E}[X^2]+2t\mathbb{E}[XY]+\mathbb{E}[Y^2]\ge 0.$$

A quadratic polynomial with positive leading coefficient that is never negative cannot have a positive discriminant: $\Delta=(2\mathbb{E}[XY])^2-4\mathbb{E}[X^2]\mathbb{E}[Y^2]\le 0\implies(\mathbb{E}[XY])^2\le\mathbb{E}[X^2]\mathbb{E}[Y^2]$. In the case of equality $\Delta=0$, so there is a t_0 with $g(t_0)=\mathbb{E}[(t_0X+Y)^2]=0$, i.e. $t_0X+Y=0$ a.s. □

Corollary 3.9 (Bounds on the Correlation)**.** $|\operatorname{Corr}(X,Y)| \le 1$*, with equality if and only if* $Y = a + bX$ *a.s. (*$b \neq 0$*).*

Proof. Apply Theorem 3.8 to the centred variables $X - \mathbb{E}X$ and $Y - \mathbb{E}Y$. □

Theorem 3.10 (Jensen's Inequality)**.** *If* $\varphi : \mathbb{R} \to \mathbb{R}$ *is convex,* $X \in L^1$ *and* $\varphi(X) \in L^1$*, then*

$$\varphi(\mathbb{E}X) \le \mathbb{E}[\varphi(X)].$$

If φ *is strictly convex, equality holds only when* X *is a.s. constant.*

Proof. Let $\mu = \mathbb{E}X$. A convex function has a supporting line at every point: $\varphi(x) \ge \varphi(\mu) + c(x-\mu)$ for all x (for some $c \in \mathbb{R}$). Setting $x = X$ and taking expectations gives $\mathbb{E}\varphi(X) \ge \varphi(\mu) + c(\mathbb{E}X - \mu) = \varphi(\mu)$. □

Theorem 3.11 (Tower Rule (Law of Total Expectation))**.** *If* $X \in L^1$*, then* $\mathbb{E}\big[\mathbb{E}[X \mid Y]\big] = \mathbb{E}[X]$*. Moreover, for every bounded measurable* h*,* $\mathbb{E}[h(Y)X \mid Y] = h(Y)\mathbb{E}[X \mid Y]$ *("taking out what is known").*

Proof (density case).

$$\mathbb{E}[m(Y)] = \int \Big(\int x f_{X|Y}(x \mid y) dx \Big) f_Y(y) dy = \iint x f_{X,Y}(x,y)\, dx\, dy = \mathbb{E}[X],$$

where the order of integration was exchanged by Fubini's theorem. The second property holds because $h(Y) = h(y)$ is constant given $Y = y$. □

Theorem 3.12 (Law of Total Variance)**.** *If* $X \in L^2$*, then*

$$\operatorname{Var}(X) = \mathbb{E}\big[\operatorname{Var}(X \mid Y)\big] + \operatorname{Var}\big(\mathbb{E}[X \mid Y]\big).$$

Proof. Let $m(Y) = \mathbb{E}[X \mid Y]$. Write $X - \mathbb{E}X = (X - m(Y)) + (m(Y) - \mathbb{E}X)$ and square. For the cross term the tower rule gives

$$\mathbb{E}\big[(X - m(Y))(m(Y) - \mathbb{E}X)\big] = \mathbb{E}\Big[(m(Y) - \mathbb{E}X)\, \mathbb{E}[X - m(Y) \mid Y]\Big] = 0,$$

because $\mathbb{E}[X - m(Y) \mid Y] = 0$. The remaining terms are $\mathbb{E}[(X - m(Y))^2] = \mathbb{E}[\operatorname{Var}(X \mid Y)]$ and $\mathbb{E}[(m(Y) - \mathbb{E}X)^2] = \operatorname{Var}(m(Y))$ (the latter uses $\mathbb{E}[m(Y)] = \mathbb{E}X$). □

Corollary 3.13 (Best Predictor)**.** *If* $X \in L^2$*, then among all measurable functions* g *of* Y*, the one minimising* $\mathbb{E}[(X - g(Y))^2]$ *is* $g(Y) = \mathbb{E}[X \mid Y]$*.*

Proof. The same decomposition as in the proof of the law of total variance gives $\mathbb{E}[(X - g(Y))^2] = \mathbb{E}[(X - m(Y))^2] + \mathbb{E}[(m(Y) - g(Y))^2]$; the second term vanishes for $g = m$. □

3.3 Worked Examples

Example 3.14 (The Binomial Distribution — the Indicator Trick)**.** $X \sim \operatorname{Bin}(n,p)$ can be written as $X = \sum_{i=1}^{n} Y_i$ with $Y_i \overset{\text{i.i.d.}}{\sim} \operatorname{Bernoulli}(p)$. By linearity $\mathbb{E}X = np$; by independence $\operatorname{Var} X = \sum \operatorname{Var} Y_i = np(1-p)$. This is much shorter than evaluating the sum $\sum_k k\binom{n}{k}p^k(1-p)^{n-k}$ directly.

Example 3.15 (Moments of the Exponential Distribution)**.** For $X \sim \operatorname{Exp}(\lambda)$, integration by parts gives $\mathbb{E}[X^k] = k!/\lambda^k$. Hence $\mathbb{E}X = 1/\lambda$ and $\operatorname{Var} X = 2/\lambda^2 - 1/\lambda^2 = 1/\lambda^2$. The MGF is $M_X(t) = \lambda/(\lambda - t)$, $t < \lambda$; its derivatives give the same moments.

Example 3.16 (A Distribution Without an Expectation: Cauchy)**.** For the standard Cauchy density $f(x) = \frac{1}{\pi(1+x^2)}$,

$$\mathbb{E}[X^+] = \int_0^\infty \frac{x}{\pi(1+x^2)}dx = \frac{1}{2\pi}\ln(1+x^2)\Big|_0^\infty = \infty,$$

and similarly $\mathbb{E}[X^-] = \infty$; that is, $\mathbb{E}X$ is *undefined*. Although the density is symmetric about zero, one cannot say "the mean is zero". In Chapters **??** and **??** this distribution will show why the assumptions of the law of large numbers and the central limit theorem are needed.

Example 3.17 (Uncorrelated but Dependent)**.** Let $X \sim \mathcal{N}(0,1)$ and $Y = X^2$. Then $\operatorname{Cov}(X,Y) = \mathbb{E}[X^3] - \mathbb{E}X\,\mathbb{E}[X^2] = 0$, yet Y is a deterministic function of X. Correlation measures only *linear* dependence. (Conversely, if (X,Y) is *jointly* normal, uncorrelatedness is equivalent to independence: Theorem **??**. Normal marginals alone are not enough: Example **??**.)

Example 3.18 (Applications of Jensen)**.**

- $\varphi(x) = x^2$: $(\mathbb{E}X)^2 \le \mathbb{E}X^2$, i.e. $\operatorname{Var} X \ge 0$.
- $\varphi(x) = 1/x$ $(x > 0)$: $\mathbb{E}[1/X] \ge 1/\mathbb{E}X$. For instance, computing an "average time" from an average speed systematically underestimates the time.
- $\varphi(x) = -\ln x$: $\mathbb{E}[\ln X] \le \ln \mathbb{E}X$, the probabilistic form of the arithmetic–geometric mean inequality. It will be used in Chapter **??** to prove the consistency of the MLE (the information inequality).
- For $S = \sqrt{S^2}$: $\mathbb{E}[S] = \mathbb{E}\sqrt{S^2} < \sqrt{\mathbb{E}S^2} = \sigma$, so S is a (downward) biased estimator of σ (Chapter **??**).

Example 3.19 (Total Variance: a Poisson–Gamma Mixture)**.** Let the annual number of claims of a policyholder be $N \mid \Lambda \sim \text{Poisson}(\Lambda)$ with risk level $\Lambda \sim \text{Gamma}(\alpha, \beta)$ (shape α, rate β). By the tower rule and the law of total variance,

$$\mathbb{E}N = \mathbb{E}[\Lambda] = \frac{\alpha}{\beta}, \qquad \operatorname{Var} N = \mathbb{E}[\Lambda] + \operatorname{Var}(\Lambda) = \frac{\alpha}{\beta} + \frac{\alpha}{\beta^2}.$$

Since $\operatorname{Var} N > \mathbb{E}N$, the mixture produces *overdispersion* relative to the Poisson. The marginal distribution of N is negative binomial. This structure is the probabilistic counterpart of the Gamma–Poisson conjugacy in Chapter **??**.

3.4 Computational Applications

3.4.1 Verifying the Cauchy–Schwarz Inequality

code/ch03/c03_1_cauchy_schwarz_check.py

```python
import numpy as np

rng = np.random.default_rng(42)
cov = [[4.0, 0.85 * 2.0 * 3.0], [0.85 * 2.0 * 3.0, 9.0]]
samples = rng.multivariate_normal([2.0, -1.0], cov, size=500_000)
X, Y = samples[:, 0], samples[:, 1]
print(f"(E[XY])^2: {(np.mean(X*Y))**2:.4f} <= E[X^2]E[Y^2]: {np.mean(X**2)*np.mean(Y**2):.4f}
    ")

# Equality case: Y = a + bX  =>  |Corr| = 1 ; equality for the centred variables
Y_lin = 3.0 - 2.0 * X
Xc, Yc = X - X.mean(), Y_lin - Y_lin.mean()
print(f"Linear relation: (E[XcYc])^2 = {np.mean(Xc*Yc)**2:.4f} , E[Xc^2]E[Yc^2] = {np.mean(Xc
    **2)*np.mean(Yc**2):.4f}")
print(f"Sample correlation (for rho=0.85): {np.corrcoef(X, Y)[0,1]:.4f}")
```

Output

```
(E[XY])^2: 9.5445 <= E[X^2]E[Y^2]: 79.7080
Linear relation: (E[XcYc])^2 = 63.7241 , E[Xc^2]E[Yc^2] = 63.7241
Sample correlation (for rho=0.85): 0.8489
```

Theoretical values: $\mathbb{E}[XY] = \mathrm{Cov} + \mathbb{E}X\mathbb{E}Y = 5.1 - 2 = 3.1$, hence $(\mathbb{E}XY)^2 = 9.61$; $\mathbb{E}X^2 = 4 + 4 = 8$, $\mathbb{E}Y^2 = 9 + 1 = 10$, product 80. The inequality is rather loose because Cauchy–Schwarz was applied to *raw* moments; in centred form the ratio is $\rho^2 = 0.7225$.

3.4.2 Uncorrelatedness versus Independence; the Law of Total Variance

code/ch03/c03_2_uncorrelated_vs_independent_total_variance.py

```python
import numpy as np

rng = np.random.default_rng(42)
X = rng.normal(size=1_000_000)
Y = X**2
print(f"Corr(X, X^2) = {np.corrcoef(X, Y)[0,1]:+.4f}   (uncorrelated)")
print(f"P(Y>1 | |X|>1) = {np.mean(Y[np.abs(X) > 1] > 1):.1f}  vs  P(Y>1) = {np.mean(Y > 1):.4f}  (dependent)")

# Poisson-Gamma mixture: Var N = E[Var(N|L)] + Var(E[N|L])
alpha, beta = 3.0, 0.5
L = rng.gamma(alpha, 1/beta, size=1_000_000)   # numpy takes scale = 1/rate
N = rng.poisson(L)
print(f"E N   : MC={N.mean():.3f}  theory={alpha/beta:.3f}")
print(f"Var N : MC={N.var():.3f}  theory={alpha/beta + alpha/beta**2:.3f}  (would be {alpha/beta:.1f} for a Poisson)")
```

Output

```
Corr(X, X^2) = -0.0006   (uncorrelated)
P(Y>1 | |X|>1) = 1.0  vs  P(Y>1) = 0.3171  (dependent)
E N   : MC=6.005  theory=6.000
Var N : MC=18.034  theory=18.000  (would be 6.0 for a Poisson)
```

3.4.3 Jensen's Inequality: $\mathbb{E}[1/X]$ and $\mathbb{E}[S]$

code/ch03/c03_3_jensen_inverse_and_s.py

```python
import numpy as np
from scipy.special import gammaln

rng = np.random.default_rng(42)
X = rng.gamma(5.0, 1.0, size=1_000_000)       # E X = 5
print(f"E[1/X] = {np.mean(1/X):.4f} >= 1/E[X] = {1/X.mean():.4f}  (theory E[1/X]=1/4)")

n, sigma = 5, 2.0
S = np.std(rng.normal(0, sigma, size=(500_000, n)), axis=1, ddof=1)
c4 = np.sqrt(2/(n-1)) * np.exp(gammaln(n/2) - gammaln((n-1)/2))
print(f"E[S] MC = {S.mean():.4f} < sigma = {sigma}  | theory c4*sigma = {c4*sigma:.4f}")
```

Output

```
E[1/X] = 0.2501 >= 1/E[X] = 0.2000  (theory E[1/X]=1/4)
E[S] MC = 1.8787 < sigma = 2.0  | theory c4*sigma = 1.8800
```

Here $c_4(n) = \sqrt{2/(n-1)}\,\Gamma(n/2)/\Gamma((n-1)/2)$ is the correction factor giving $\mathbb{E}[S] = c_4\sigma$ for a normal sample ($c_4 \approx 0.940$ for $n = 5$).

3.5 Comments and Notes

Remark 3.20 (Numerical Stability When Computing a Variance). The formula $\operatorname{Var} X = \mathbb{E}X^2 - (\mathbb{E}X)^2$ is correct in theory but numerically dangerous: if the mean is large relative to the standard deviation, one subtracts two large numbers and floating-point errors are magnified (catastrophic cancellation). In practice a two-pass algorithm or Welford's one-pass algorithm should be used; `np.var` uses the two-pass method.

Remark 3.21 (Hierarchy of Moment Existence). For $1 \le p < q$, $\mathbb{E}|X|^q < \infty \implies \mathbb{E}|X|^p < \infty$ (Lyapunov's inequality $(\mathbb{E}|X|^p)^{1/p} \le (\mathbb{E}|X|^q)^{1/q}$, a consequence of Jensen). For example, the t_ν distribution has moments only of order less than ν; t_2 has a mean but infinite variance.

Remark 3.22 (Links Ahead). The Cauchy–Schwarz inequality is the key step for the Cramér–Rao lower bound in Chapter **??**; the law of total variance for the Rao–Blackwell theorem in Chapter **??**; and Jensen's inequality for the proof of the consistency of the MLE in Chapter **??**.

3.6 Chapter Summary

- The expectation is a Lebesgue integral (Definition 3.1). It is linear and monotone, and multiplicative for independent variables (Proposition 3.7).
- The Cauchy–Schwarz inequality (Theorem 3.8) yields $|\operatorname{Corr}(X, Y)| \le 1$, with equality only for a linear relation (Corollary 3.9).
- Jensen's inequality: if φ is convex, $\varphi(\mathbb{E}X) \le \mathbb{E}\varphi(X)$ (Theorem 3.10). The information inequality and the Rao–Blackwell theorem are applications of it.
- The tower rule $\mathbb{E}[\mathbb{E}(X \mid Y)] = \mathbb{E}X$ (Theorem 3.11) and the law of total variance $\operatorname{Var} X = \mathbb{E}\operatorname{Var}(X \mid Y) + \operatorname{Var}\mathbb{E}(X \mid Y)$ (Theorem 3.12) simplify moment calculations in hierarchical models.
- $\mathbb{E}(X \mid Y)$ is the predictor that minimises the mean squared error among functions of Y (Corollary 3.13). This is the basic idea behind regression and Rao–Blackwell.
- The moment generating function produces moments through its derivatives. When it is not defined in a neighbourhood of zero, the characteristic function of Chapter **??** is used instead.

Further reading. [16, Chapter 2], [81, Chapter 3]; for expectation as a Lebesgue integral see [22, Sections 1.5–1.6].

3.7 Exercises

Exercise 3.1. Show using Fubini's theorem that $\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t)\,dt$ for $X \ge 0$. Use this to find $\mathbb{E}X = 1/p$ for $X \sim \text{Geometric}(p)$ (on $\{1, 2, \dots\}$). → Solution

Exercise 3.2. Let $X \sim \text{Uniform}(0, 1)$ and $Y \mid X = x \sim \text{Uniform}(0, x)$. Compute $\mathbb{E}Y$ and $\operatorname{Var} Y$ using the tower rule and the law of total variance, and verify the result by Monte Carlo. → Solution

Exercise 3.3. Show that $\operatorname{Var}(X) = \min_{c \in \mathbb{R}} \mathbb{E}[(X - c)^2]$ and $\mathbb{E}|X - m| = \min_c \mathbb{E}|X - c|$ (m a median). How do these two results relate to the Bayes estimators of Chapter **??**? → Solution

Exercise 3.4. Let $Z_1, Z_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$, $X = Z_1$ and $Y = \rho Z_1 + \sqrt{1 - \rho^2} Z_2$. Show that $\operatorname{Corr}(X, Y) = \rho$, $\mathbb{E}[Y \mid X] = \rho X$ and $\operatorname{Var}(Y \mid X) = 1 - \rho^2$. → Solution

Part II

Appendices

Appendix A

Solutions to the Exercises

This appendix contains solutions to all the exercises in the book. The heading of each solution links to the exercise, and the "→ Solution" link at the end of each exercise leads here. Where an exercise asks for numerical values or a simulation, the values given were obtained by actual computation; simulation results may differ in the last digits depending on the random seed. The same libraries as in the chapters (`numpy`, `scipy`) are used; short code is given only where it is part of the solution.

The solutions are kept as short as possible, but each step names the result in the book on which it rests. When the claim of an exercise holds only under certain conditions (e.g. Exercise **??**), the solution states those conditions explicitly.

A.1 Probability Spaces

Chapter 1.

Exercise 1.1. Let $\mathcal{F} = \mathcal{F}_1 \cap \mathcal{F}_2$. Since Ω is in both, $\Omega \in \mathcal{F}$. If $A \in \mathcal{F}$, then A belongs to both families, and since both are closed under complements, $A^c \in \mathcal{F}_1 \cap \mathcal{F}_2$. The same argument works for countable unions. (The same proof works for the intersection of arbitrarily many σ-algebras; the definition of the generated σ-algebra rests on this.) Counterexample for the union: $\Omega = \{1,2,3\}$, $\mathcal{F}_1 = \{\emptyset, \{1\}, \{2,3\}, \Omega\}$, $\mathcal{F}_2 = \{\emptyset, \{2\}, \{1,3\}, \Omega\}$. Then $\{1\}, \{2\} \in \mathcal{F}_1 \cup \mathcal{F}_2$, but $\{1\} \cup \{2\} = \{1,2\}$ is in neither; the union is not closed under unions.

Exercise 1.2. Applying the two-event formula to $A \cup B$ and C and using distributivity gives

$$\mathbb{P}(A \cup B \cup C) = \mathbb{P}(A) + \mathbb{P}(B) + \mathbb{P}(C) - \mathbb{P}(AB) - \mathbb{P}(AC) - \mathbb{P}(BC) + \mathbb{P}(ABC)$$

($AB = A \cap B$). The general formula

$$\mathbb{P}\Big(\bigcup_{i=1}^{n} A_i\Big) = \sum_{k=1}^{n} (-1)^{k+1} \sum_{i_1 < \cdots < i_k} \mathbb{P}(A_{i_1} \cap \cdots \cap A_{i_k})$$

follows by induction: for $B = \bigcup_{i \le n} A_i$, $\mathbb{P}(B \cup A_{n+1}) = \mathbb{P}(B) + \mathbb{P}(A_{n+1}) - \mathbb{P}\big(\bigcup_{i \le n}(A_i \cap A_{n+1})\big)$; apply the induction hypothesis to the two unions on the right and collect terms. *Matching problem:* let $A_i = \{\text{letter } i \text{ is in the right envelope}\}$. The probability that k given letters are in the right place is $(n-k)!/n!$, and there are $\binom{n}{k}$ such choices; the inner sum equals $1/k!$. The probability that no letter is in the right place is

$$1 - \sum_{k=1}^{n} \frac{(-1)^{k+1}}{k!} = \sum_{k=0}^{n} \frac{(-1)^k}{k!} \xrightarrow{n\to\infty} e^{-1} \approx 0.368.$$

Convergence is very fast: for $n = 6$ the difference is below 10^{-4}.

Exercise 1.3. By De Morgan and Boole's inequality, $\mathbb{P}(\bigcap A_i) = 1 - \mathbb{P}(\bigcup A_i^c) \geq 1 - \sum \mathbb{P}(A_i^c)$. *Multiple testing:* for m true null hypotheses let $A_i = \{\text{test } i \text{ does not reject falsely}\}$. If each test is carried out at level α/m, then $\mathbb{P}(A_i^c) \leq \alpha/m$, and the probability of no false rejection at all is $\geq 1 - \alpha$. This is the Bonferroni correction (see Exercise **??**); it holds whatever the dependence between the tests, but it is conservative when the tests are positively dependent.

Exercise 1.4. $\mathbb{P}(A \cap B^c) = \mathbb{P}(A) - \mathbb{P}(A \cap B) = \mathbb{P}(A) - \mathbb{P}(A)\mathbb{P}(B) = \mathbb{P}(A)\mathbb{P}(B^c)$, so $A \perp\!\!\!\perp B^c$. Applying this result again to the pair (B^c, A) gives $B^c \perp\!\!\!\perp A^c$.

Exercise 1.5. With prevalence π the positive predictive value is $\mathrm{PPV}(\pi) = \frac{0.99\pi}{0.99\pi + 0.05(1-\pi)}$. Then $\mathrm{PPV} > \frac{1}{2} \iff 0.99\pi > 0.05(1-\pi) \iff \pi > \frac{0.05}{1.04} \approx 0.0481$. So unless the prevalence exceeds about 4.8%, most positive results are false positives. For the plot:

```
import numpy as np, matplotlib.pyplot as plt
pi = np.linspace(0.001, 0.5, 500)
ppv = 0.99*pi / (0.99*pi + 0.05*(1 - pi))
plt.semilogx(pi, ppv); plt.axhline(0.5, ls=":"); plt.axvline(0.05/1.04, ls=":")
plt.xlabel("prevalence"); plt.ylabel("P(D | +)"); plt.show()
```

A.2 Random Variables and Distribution Functions

Chapter 2.

Exercise 2.1. $g(x) = -2\ln x$ is strictly decreasing on $(0,1)$; its inverse is $x = e^{-y/2}$ with $|dx/dy| = \frac{1}{2}e^{-y/2}$. By Theorem 2.16, $f_Y(y) = 1 \cdot \frac{1}{2}e^{-y/2}$, $y > 0$: this is $\mathrm{Exp}(\frac{1}{2}) = \mathrm{Gamma}(1, \frac{1}{2}) = \chi^2_2$. *Fisher's method:* the continuous p-values of m independent tests are $\mathrm{Uniform}(0,1)$ under H_0 (Theorem **??**); hence $-2\sum_{i=1}^{m} \ln p_i \sim \chi^2_{2m}$, and large values of this statistic are evidence that "at least one null hypothesis is false".

Exercise 2.2. $u = F(x) = 1 - e^{-x^k} \iff x = F^{-1}(u) = \big(-\ln(1-u)\big)^{1/k}$. Mean: $\mathbb{E}X = \int_0^\infty k x^k e^{-x^k} dx$; with $t = x^k$ this is $\int_0^\infty t^{1/k} e^{-t} dt = \Gamma(1 + 1/k)$. Simulation with 4×10^5 draws: 1.994 for $k = 0.5$ (theory 2), 0.9031 for $k = 1.5$ (0.9027), 0.8934 for $k = 3$ (0.8930).

Exercise 2.3. $\{\max(X,Y) \leq z\} = \{X \leq z\} \cap \{Y \leq z\}$, so by independence $F_{\max}(z) = F(z)G(z)$. $\{\min(X,Y) > z\} = \{X > z\} \cap \{Y > z\}$, hence $F_{\min}(z) = 1 - [1 - F(z)][1 - G(z)]$. By induction, $\mathbb{P}(X_{(n)} \leq x) = \prod_i \mathbb{P}(X_i \leq x) = F(x)^n$.

Exercise 2.4. $2e^{-2n(0.02)^2} \leq 0.05 \iff n \geq \frac{\ln 40}{2 \cdot 0.0004} \approx 4611.1$; the smallest n is 4612. In general, with $\varepsilon_n = \sqrt{\ln(2/\alpha)/(2n)}$, the band

$$L(x) = \max\{\hat{F}_n(x) - \varepsilon_n, 0\}, \qquad U(x) = \min\{\hat{F}_n(x) + \varepsilon_n, 1\}$$

satisfies $\mathbb{P}(L(x) \leq F(x) \leq U(x)\ \forall x) \geq 1 - \alpha$. This is a simultaneous, distribution-free confidence band valid for *all* x.

A.3 Expectation and Variance

Chapter 3.

Exercise 3.1. $X(\omega) = \int_0^\infty \mathbb{I}\{X(\omega) > t\} dt$. Since the integrand is non-negative, Fubini–Tonelli gives $\mathbb{E}X = \int_0^\infty \mathbb{E}\,\mathbb{I}\{X > t\}\, dt = \int_0^\infty \mathbb{P}(X > t)\, dt$. For the geometric distribution $\mathbb{P}(X > t) = (1-p)^{\lfloor t \rfloor}$ is a step function, and the integral is $\sum_{k=0}^{\infty} (1-p)^k = 1/p$.

Exercise 3.2. $\mathbb{E}[Y \mid X] = X/2$ and $\mathrm{Var}(Y \mid X) = X^2/12$. Tower rule: $\mathbb{E}Y = \mathbb{E}[X/2] = \frac{1}{4}$. Total variance: $\mathrm{Var}\, Y = \mathbb{E}[X^2/12] + \mathrm{Var}(X/2) = \frac{1}{36} + \frac{1}{48} = \frac{7}{144} \approx 0.04861$. Monte Carlo with 10^6 draws: $\mathbb{E}Y \approx 0.2499$, $\mathrm{Var}\, Y \approx 0.04860$.

Exercise 3.3. $\mathbb{E}(X-c)^2 = \operatorname{Var} X + (\mathbb{E}X - c)^2$ is minimised at $c = \mathbb{E}X$, with minimum $\operatorname{Var} X$. $g(c) = \mathbb{E}|X-c|$ is convex with right derivative $g'_+(c) = \mathbb{P}(X \le c) - \mathbb{P}(X > c) = 2F(c) - 1$, which is ≤ 0 for $c < m$ and ≥ 0 for $c \ge m$; hence g is minimised at the median. *Bayes connection:* applying the same computations to the posterior distribution, the Bayes estimator is the posterior mean under squared loss and the posterior median under absolute loss (Theorem **??**).

Exercise 3.4. $\operatorname{Var} Y = \rho^2 + (1-\rho^2) = 1$ and $\operatorname{Cov}(X,Y) = \rho \operatorname{Var} Z_1 = \rho$, so Corr $= \rho$. Since (X,Y) is a linear transformation of (Z_1, Z_2), it is jointly normal. $Y - \rho X = \sqrt{1-\rho^2}Z_2$ is independent of $X = Z_1$; therefore $\mathbb{E}[Y \mid X] = \rho X + \mathbb{E}[\sqrt{1-\rho^2}Z_2] = \rho X$ and $\operatorname{Var}(Y \mid X) = 1 - \rho^2$. (A special case of Theorem **??**(4); this construction is also the standard way of generating a bivariate normal.)

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